

# REPORT:

## *A3.2.5. – Documented example of wettability test protocol*

### *Work package 3*

This report was written as part of activity 3.2.5 from the EMPIR Establishing Metrology Standards in Microfluidic Devices (MFMET) project. The three-year European project commenced on 1<sup>st</sup> June 2021 and focused on providing a generic methodology of accurate measurement of a particular quantity in a microfluidic device by utilising standardised methods and reference documents, e.g. VIM & GUM. For more details about this project, please visit [www.mfmet.eu](http://www.mfmet.eu)

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## Contents

|                                                                                               |    |
|-----------------------------------------------------------------------------------------------|----|
| 1. Scope                                                                                      | 3  |
| 2. Introduction                                                                               | 3  |
| 3. Documented example                                                                         | 4  |
| Measurement setup                                                                             | 4  |
| Measurement setup calibration                                                                 | 5  |
| Owens-Wendt-Rabel & Kaelble model                                                             | 9  |
| Application of the OWRK model for experimental assessment of surface energy of D263bio sample | 11 |
| Conclusion                                                                                    | 14 |

## 1. Scope

This document provides information and procedures to test the hydrophobicity, hydrophilicity and wettability, focused on microfluidic device applications, ensuring traceability to national standards and conformity to existing normative standards, provided these exist.

| Activity number | Activity description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Partners<br>(Lead in bold)                             |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| A3.2.5<br>M17   | IMTAG, with support from UofG, CETIAT, EnablingMNT, IPQ and TUBITAK will perform at IMTAG facility a documented example of the test protocol developed in A3.2.2 (i.e. the documented example will apply the test protocol to a specific example, such as wettability.<br>IMTAG with support from UofG, CETIAT, EnablingMNT, IPQ and TUBITAK will write a technical report on the documented example of the test protocol for hydrophobicity/hydrophilicity, with a focus on microfluidic device applications. | <b>IMTAG</b> , UofG, CETIAT, EnablingMNT, IPQ, TUBITAK |

## 2. Introduction

CETIAT used the protocol developed in activity A3.2.2 to perform a **surface wettability** assessment of a glass material (provided by IMT, referenced D263bio), of use in typical microfluidics application. This report is intended to be used as a follow-up of A3.2.2 report, describe in details the measurand (contact angle) and the property assessed (wettability quantified by the surface energy of a given material).

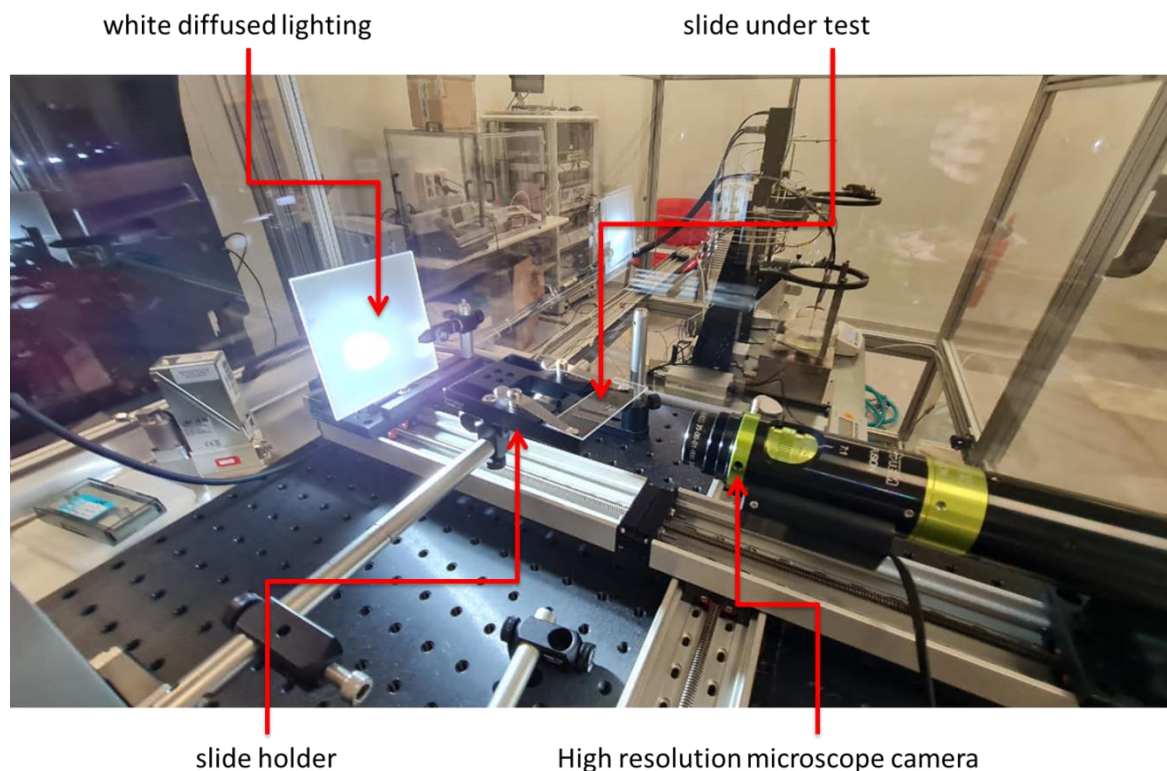
### 3. Documented example

#### Measurement setup

The following conditions described in A3.2.2 test protocol to measure contact angles on flat has been followed for this documented example:

- Setting up the contact angle measuring system should be free of vibrations, intense air flow, and intense exposure to light from the outside. The system should be aligned horizontally.
- Test conditions should be carried out at  $23 \pm 2^\circ\text{C}$  and a relative humidity at  $50 \pm 5\%$ , the test media should have the same temperature.
- The sample specimen should be placed on a sample holder, its height should be positioned in the lower half of the image and horizontally aligned.
- The chosen test liquid should be prepared and filled in a syringe without contamination or bubbles.
- The needle/cannula is moved in the upper margin of the image; focus, contrast and brightness should be adjusted. The magnification should be adjusted in order to have the width of the contour of the drop to take up 2/3 of the measurement image.

The following picture shows the measurement setup:



*Figure 1: picture of the measurement setup.*

The material under was a clean glass slide of 1 mm thickness, 25 mm by 75 mm length, provided in a sealed box, as shown in the following picture:

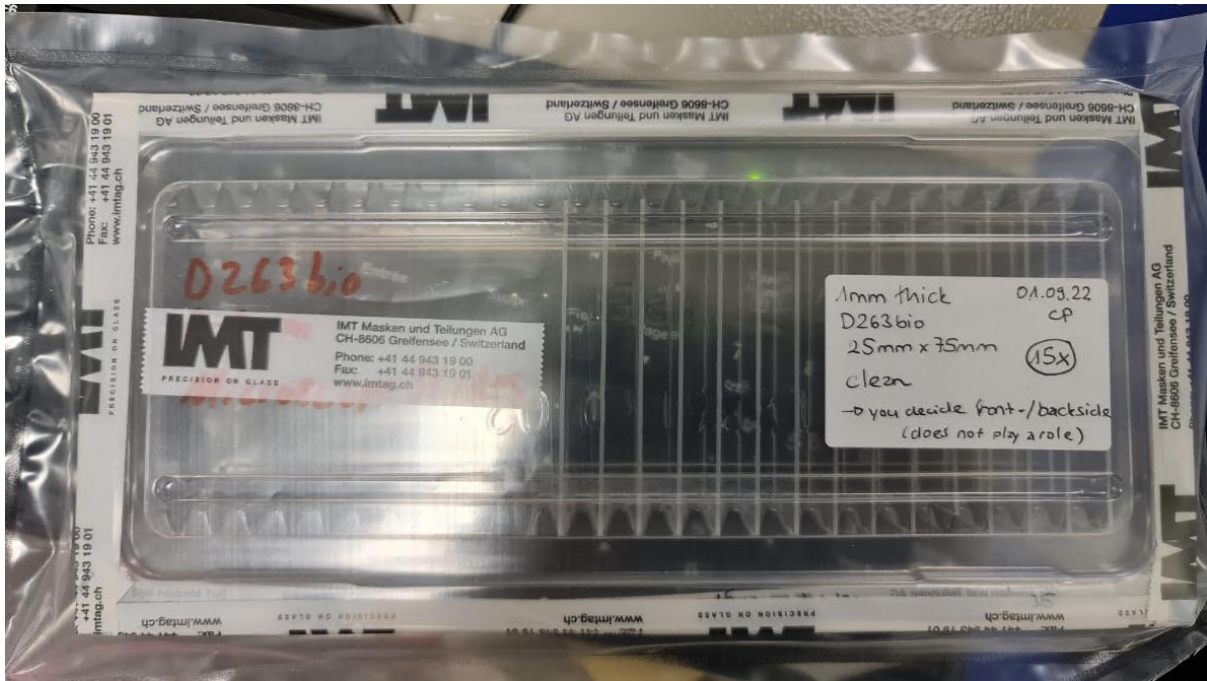


Figure 2: picture of the sealed box containing the tested D263bio glass slides

### Measurement setup calibration

Shape standard samples CP24 have been used to calibrate the Young-Laplace method for measuring the contact angle. These standard samples allow the contact angle measurement to be calibrated with an accuracy of  $0.1^\circ$ . The set CP24 additionally includes a certificate which confirms the high accuracy of the print. The angle standard itself were calibrated and provided with a calibration certificate by a ISO17025 laboratory (KRÜSS, DAkkS accredited in this example). Three standards of  $30^\circ$ ,  $60^\circ$  and  $120^\circ$  angle have been used.

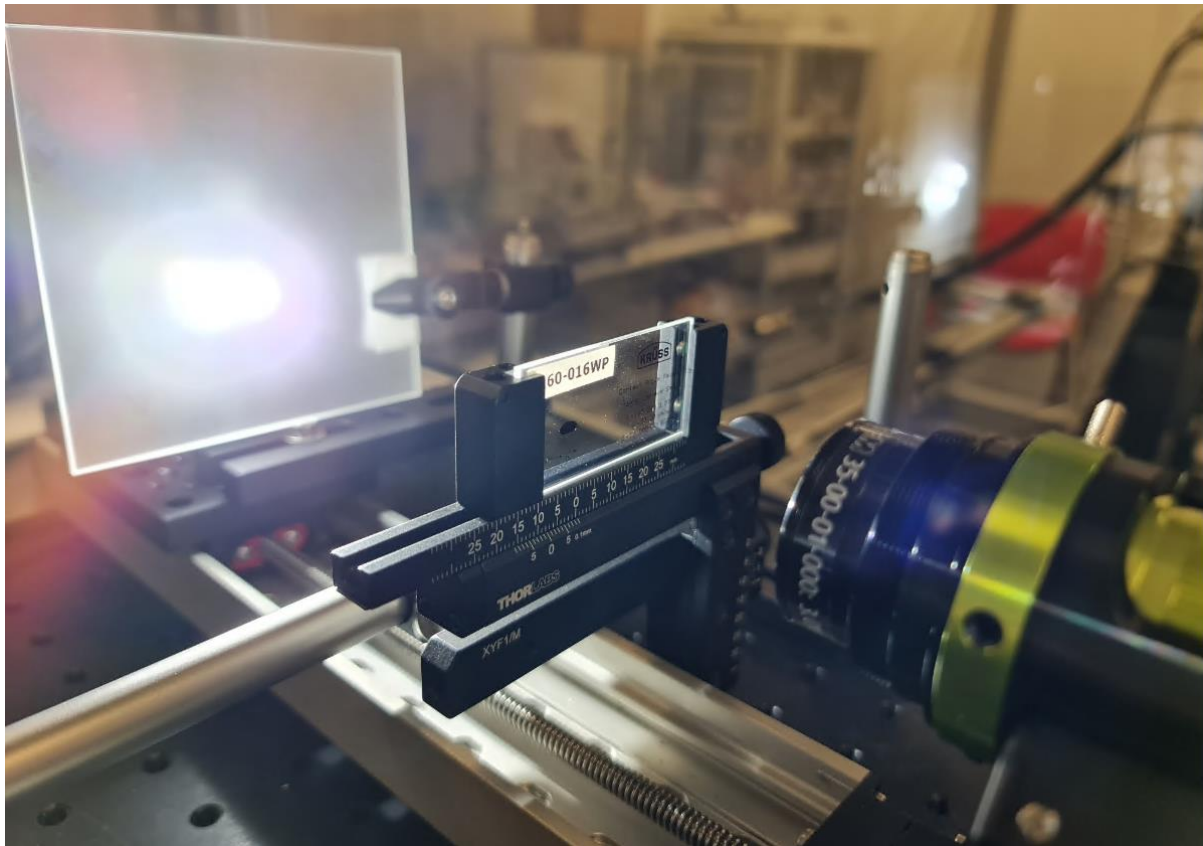
The following picture shows the  $120^\circ$  angle standard, with a  $0.04^\circ$  calibration uncertainty ( $k=2$ ).



Figure 3: KRÜSS CP24  $120^\circ$  angle standard



The following picture shows the KRUSS CP24 60° angle standard under use during the calibration of the setup.

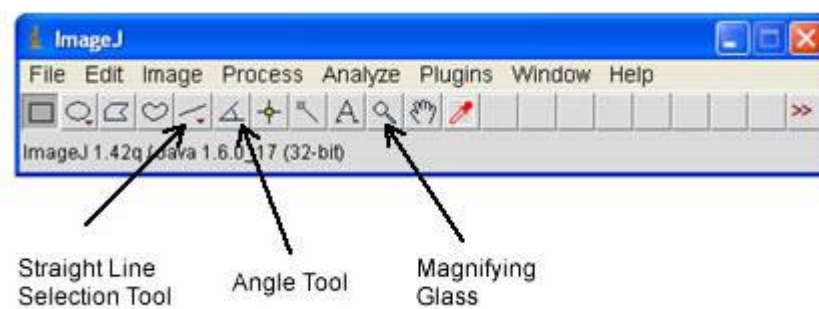


*Figure 4: KRUSS CP24 60° angle standard under use during the calibration of the setup*

For the calibration as well as for the contact angle measurement, the ImageJ Fiji software (free and open source, available at <https://imagej.net/software/fiji/downloads>) has been used. Fiji has the advantage to be provided as a distribution of ImageJ with plugins, including contact angle measurements plugins.

Three tools have been used for the calibration the contact angle measurements by Fiji software:

- "manual" angle tool, directly available in ImageJ toolbar:



*Figure 5: angle measurement tool in ImageJ toolbar*

The following picture shows an example of contact angle measurement using this tool, performed for this documented example.

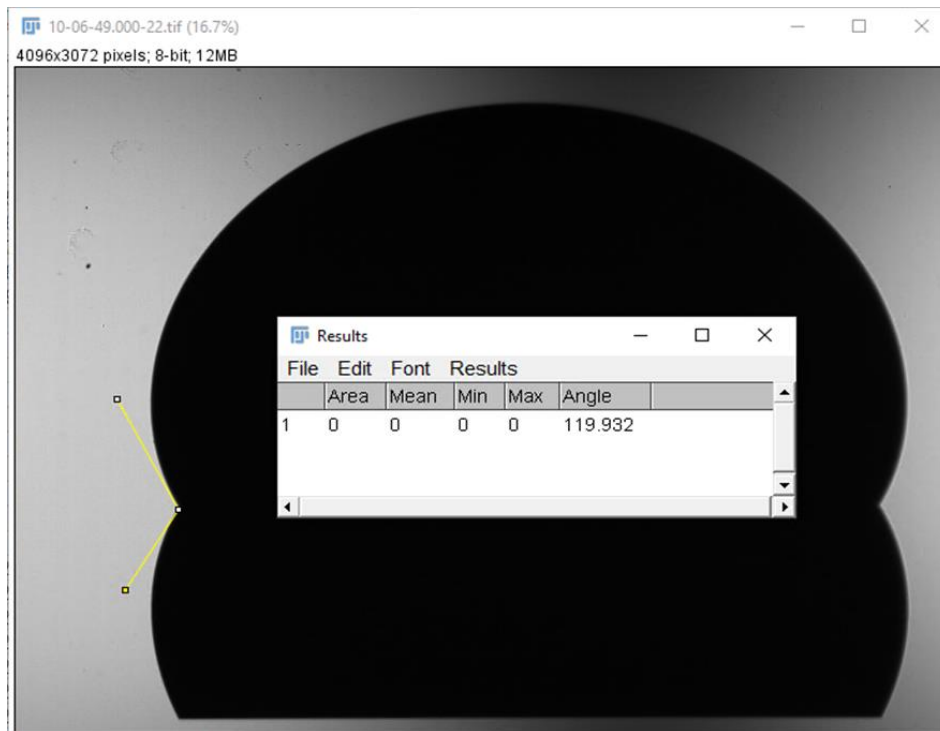


Figure 6: manual contact angle measurement of a 120° angle standard

- "DropSnake" Drop Shape Analysis plugin, developed and provided by EPFL, available at <http://bigwww.epfl.ch/demo/dropanalysis/>  
The following picture shows an example of contact angle measurement using this tool, performed for this documented example:

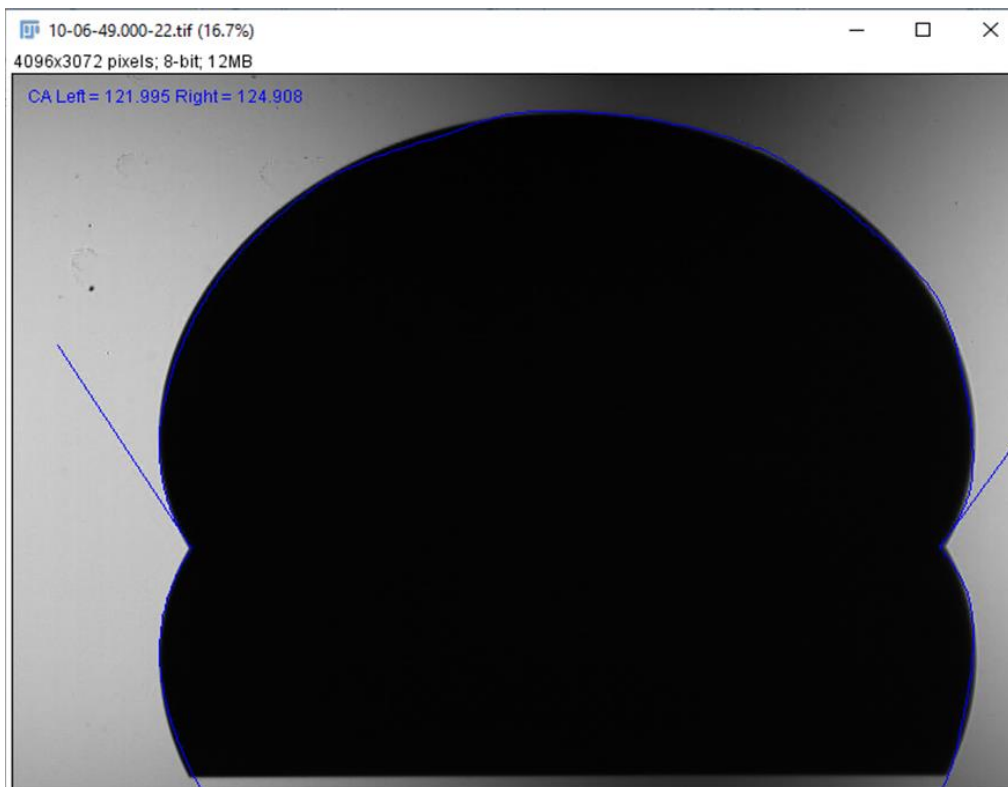


Figure 7: DropSnake contact angle measurement of a 120° standard



- "LB-ADSA" (Low Bond Axisymmetric Drop Shape Analysis) Drop Shape Analysis plugin, developed and provided by EPFL, also available at <http://bigwww.epfl.ch/demo/dropanalysis/>  
The following picture shows an example of contact angle measurement using this tool, performed for this documented example:

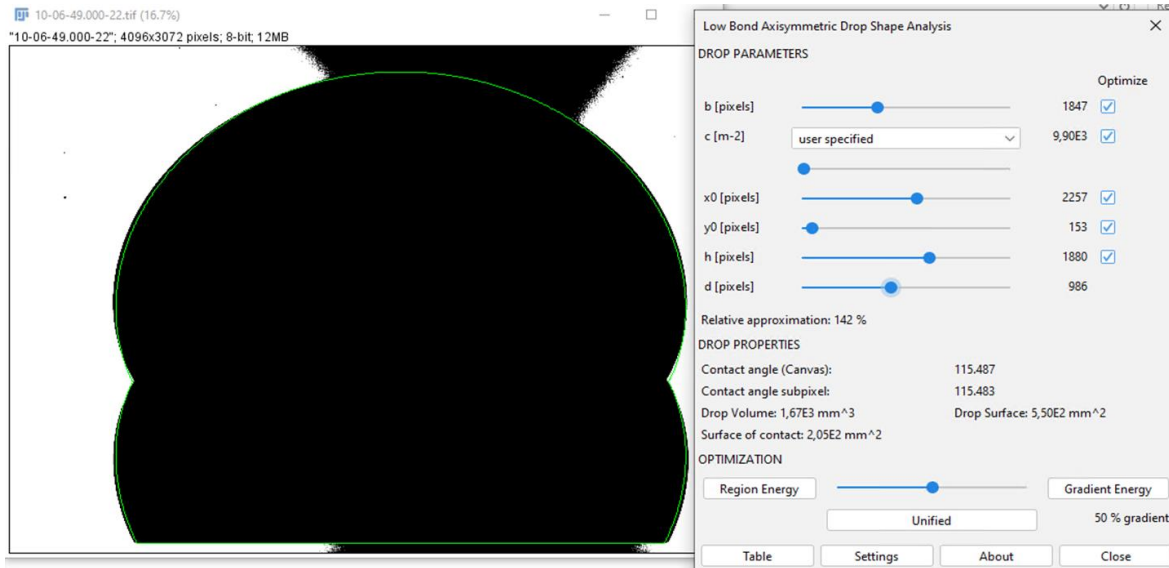


Figure 8: LB-ADSA contact angle measurement of a 120° angle standard

More information on the plugins algorithms and functionalities are available on the website above.

The following table shows the calibration results using the three tools (manual, DropSnake, LB-ADSA) for a 120° calibrated KRUS CP24 angle standard with a 0.04° (k=2) calibration uncertainty.

| Angle (°) | Manual  | DropSnake | LB-ADSA |
|-----------|---------|-----------|---------|
| 120       | 119.932 | 124.908   | 115.487 |

From the table above, it can be clearly stated that the most accurate method for contact angle measurement remains the manual method, which gives an overall accuracy of much better accuracy compared to automatic or semi-automatic plugins when calibrated against a traceable standard of 0.04° (k=2) uncertainty.

Therefore, we decided to use the manual tool for the complete calibration of the measurement setup and all contact angle measurements stated in this report.

The following table shows the calibration results for all angle standards:

| Angle (°)                        | Manual (°) | Error (°) |
|----------------------------------|------------|-----------|
| 30                               | 30.554     | 0.554     |
| 30                               | 29.346     | -0.654    |
| 30                               | 29.774     | -0.226    |
| 60                               | 59.837     | -0.163    |
| 60                               | 59.701     | -0.299    |
| 60                               | 59.320     | -0.68     |
| 120                              | 119.932    | -0.068    |
| 120                              | 120.201    | 0.201     |
| 120                              | 120.132    | 0.132     |
| Average error                    |            | -0.133    |
| Standard deviation of the errors |            | 0.398     |

From the results presented in the table above, we calculate an overall contact angle measurement uncertainty of 1° for the manual method when calibrated using traceable standards, as detailed in the equation below:

$$U(k = 2)_{\text{contact angle}} = 2 * \sqrt{\left(\frac{U_{\text{angle standard}}}{2}\right)^2 + (\text{standard deviation})^2 + |\text{average error}|}$$

### Owens-Wendt-Rabel & Kaelble model

In order to calculate the surface energy of the D263bio glass sample, we chose to use the Owens-Wendt-Rabel & Kaelble method, given the recommendations of A3.2.2, the literature review, and the experience of CETIAT. The following paragraphs are extracted from <https://www.ossila.com/en-eu/pages/a-guide-to-surface-energy> and summarized the method used for this example.

The Owens-Wendt-Rabel & Kaelble model (OWRK) was published in 1969 and 1970. It is mathematically equivalent to the Fowkes model, but derived from different principles.

The OWRK model is shown in the following equation:

$$\frac{\sigma_l(\cos\theta + 1)}{2(\sqrt{\sigma_l^D})} = \left(\sqrt{\sigma_s^P}\right) \frac{\sqrt{\sigma_l^P}}{\sqrt{\sigma_l^D}} + \sqrt{\sigma_s^D}$$

Figure 9: Owens-Wendt-Rabel & Kaelble model, which is mathematically equivalent to the Fowkes equation, but has been expressed in the form  $y = mx + c$ .

OWRK requires at least two liquids with known dispersive and polar interactions. If these are not known they can be determined with a reference solid (such as untreated PTFE). This can be assumed to have a surface energy of ~18.0 mN/m and be incapable of polar interactions, meaning  $\sigma_s = \sigma_s^D = 18$  and  $\sigma_s^P = 0$ . By inserting this into the previous equation, we can achieve the following:

$$\sigma_l^D = \frac{\sigma_l^2 (\cos \theta_{PTFE} + 1)^2}{72}$$

*Figure 10: A modification of the OWRK model that can be used to calculate the dispersive interaction of liquids from their contact angle on PTFE.*

As a result, a contact angle on PTFE can be used to calculate  $\sigma_l^D$  for any liquid where the overall surface tension is known.  $\sigma_l^P$  can then be calculated using the difference between the overall surface tension and  $\sigma_l^D$ .

Once dispersive and polar interactions of the liquids are known, they can be used to calculate surface energy of a new surface, by plotting each liquid on a graph in the form of the OWRK model. The slope between the points of the liquids will equate to  $\sqrt{\sigma_s^P}$ , and the intercept will equate to  $\sqrt{\sigma_s^D}$ . The overall surface energy can then be calculated using the following:

$$\sigma_s = \sigma_s^D + \sigma_s^P$$

*Figure 11: Surface energy for Fowkes theory is a sum of dispersive and polar components. The same is true for surface tension of the liquid (not shown)*

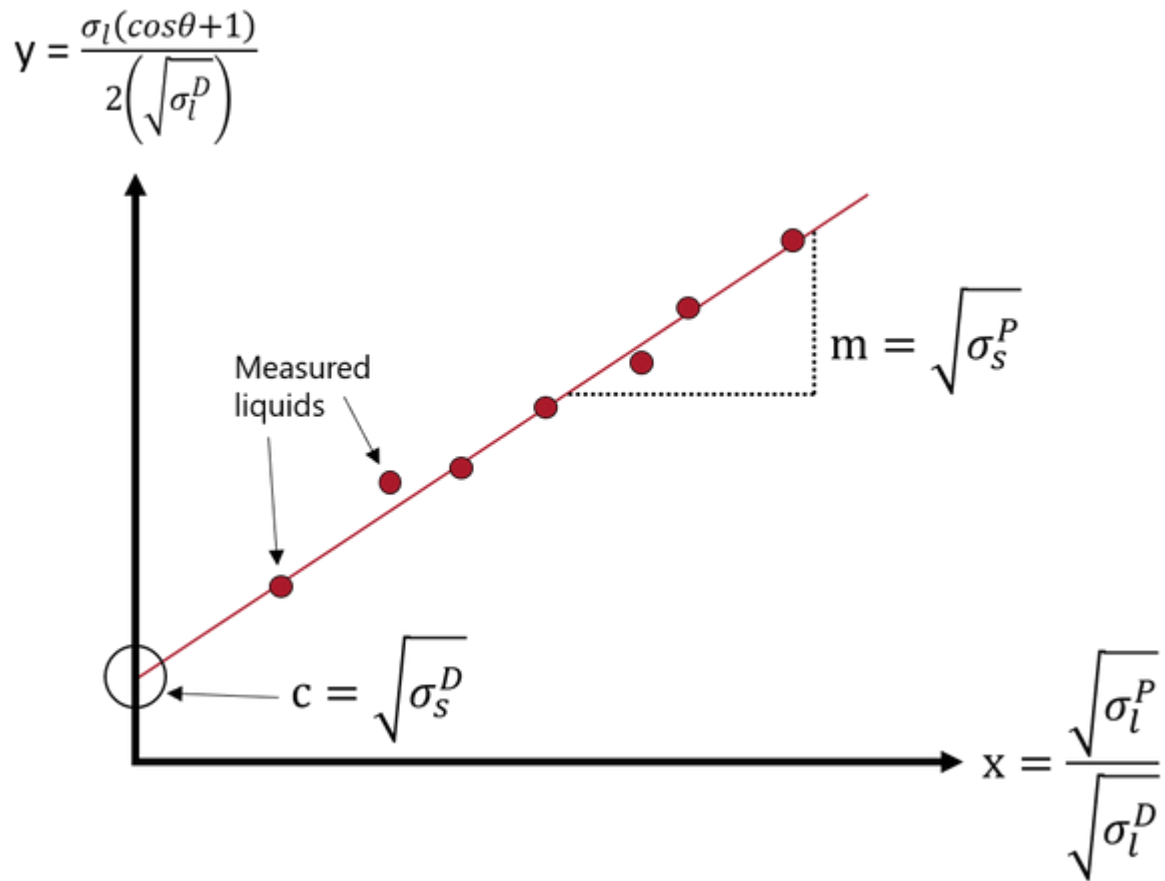


Figure 12: An OWRK plot, where contact angle is plotted against surface tension in the form of Eq.9, and the components of surface energy are found from the intercept and gradient of a best fit line.

OWRK can be used for similar materials to Fowkes, but fits better to slightly lower energy surfaces, and requires much more experimental work.

### Application of the OWRK model for experimental assessment of surface energy of D263bio sample

For this documented example, we used the OWRK model and method described to assess the surface energy of a D263bio sample using three liquids of known surface tension, dispersive fraction, and polar fraction, given in the ISO 19403-2:2017 standard, as shown in the table below.

**Table 1 — Suggested test liquids**

| Test liquid                                                                                                                                                                                                                                                                                                                                                                                                                | Surface tension<br>$\sigma_l$<br>mN/m | Dispersive fraction<br>$\sigma_l^d$<br>mN/m | Polar fraction<br>$\sigma_l^p$<br>mN/m | Source        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------|----------------------------------------|---------------|
| Water                                                                                                                                                                                                                                                                                                                                                                                                                      | 72,8                                  | 21,8                                        | 51,0                                   | Reference [4] |
| Di-iodomethane <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                | 50,8                                  | 50,8                                        | 0,0                                    | Reference [4] |
| 1,2-ethanediol<br>(ethylene glycol)                                                                                                                                                                                                                                                                                                                                                                                        | 47,7                                  | 30,9                                        | 16,8                                   | Reference [4] |
| 1,2,3-propanetriol<br>(glycerol)                                                                                                                                                                                                                                                                                                                                                                                           | 63,4                                  | 37,0                                        | 26,4                                   | Reference [4] |
| Hexadecane                                                                                                                                                                                                                                                                                                                                                                                                                 | 27,6                                  | 27,6                                        | 0,0                                    | Reference [4] |
| 1- bromo-<br>naphthalene <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                      | 44,6                                  | 44,6                                        | 0,0                                    | Reference [4] |
| Benzyl alcohol                                                                                                                                                                                                                                                                                                                                                                                                             | 38,9                                  | 29,0                                        | 9,9                                    | Reference [5] |
| Decalin<br>(isomer mixture)                                                                                                                                                                                                                                                                                                                                                                                                | 30,6                                  | 30,6                                        | 0,0                                    | Reference [4] |
| cis-Decalin                                                                                                                                                                                                                                                                                                                                                                                                                | 32,2                                  | 32,2                                        | 0,0                                    | Reference [2] |
| trans-Decalin                                                                                                                                                                                                                                                                                                                                                                                                              | 29,9                                  | 29,9                                        | 0,0                                    | Reference [2] |
| <sup>a</sup> Di-iodomethane is relative instable, yellowing after short time by splitting-off iodine.<br>Di-iodomethane swells and dissolves a lot of plastics and of organic coatings.<br>Di-iodomethane reacts with common metals (e.g. magnesium).<br><sup>b</sup> 1-bromo-naphthalene reacts with common metals (e.g. magnesium).<br>1-bromo-naphthalene tends to swelling and dissolving of high-molecular compounds. |                                       |                                             |                                        |               |

*Figure 13: extract for ISO 19403-2:2017*

For this documented example, we chose ultra pure water (ISO class 3), di-iodomethane, and ethylene glycol, as these liquids have different polar and dispersive fractions, and gives distinctive points in the OWRK plot.

For each test liquids, we cleaned the sample using dry wipes, and repeated the contact angle measurement three times using the manual angle measurement tool of ImageJ.

The following table shows the results obtained;

| experimental test<br>liquid contact<br>angles (°) | water | di-iodomethane | ethylene glycol |
|---------------------------------------------------|-------|----------------|-----------------|
|                                                   | 26.57 | 38.04          | 41.19           |
|                                                   | 21.56 | 34.02          | 33.22           |
|                                                   | 20.66 | 41.09          | 34.89           |
| Average                                           | 22.93 | 37.71          | 34.89           |

For each average contact angle, we calculated the cosine of the angle  $\cos(\theta)$ , the x and y given by the equations recalled in the OWRK plot presented in figure 12.

The following table shows the values obtained.

| test liquid     | theta | cos(theta) | sig d l | sig p l | sig l | y     | x    |
|-----------------|-------|------------|---------|---------|-------|-------|------|
| water           | 22.93 | 0.92       | 21.8    | 51      | 72.8  | 14.98 | 1.53 |
| diiodomethane   | 37.71 | 0.79       | 50.8    | 0       | 50.8  | 6.38  | 0.00 |
| ethylene glycol | 34.89 | 0.82       | 30.9    | 16.8    | 47.7  | 7.81  | 0.74 |

The x and y calculated allows for drawing the following experimental OWRK plot:

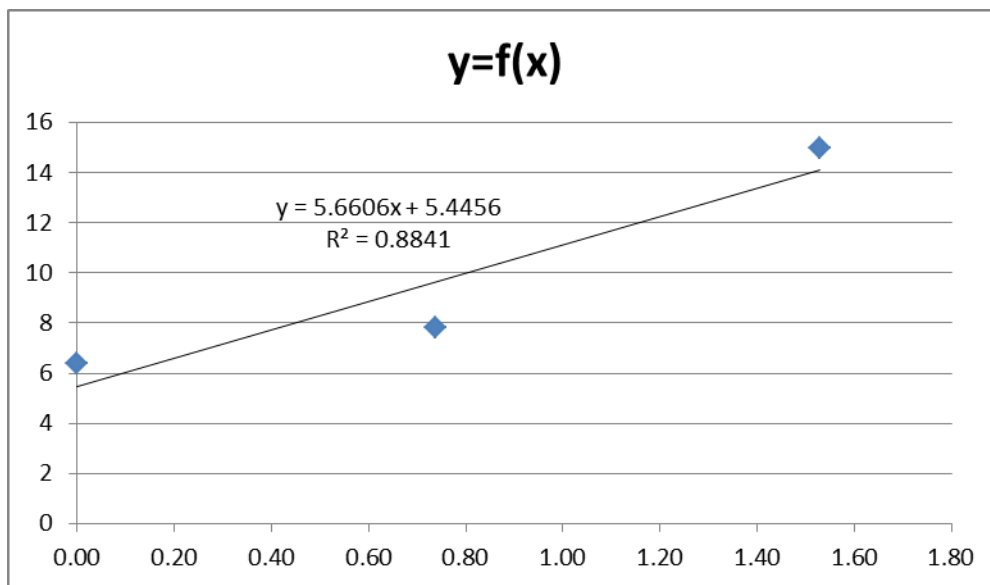


Figure 14: experimental OWRK plot for a D263bio sample tested with ultra-pur water, di-iodomethane, and ethylene glycol

A linear regression on the plot above allows obtaining:

- the polar fraction  $m$  as the square of the slope of the linear fit
- the dispersive fraction  $c$  as the square of the y-intercept of the linear fit

The numerical values obtained are the following:

|       |       |      |           |
|-------|-------|------|-----------|
| $m$   | 5.66  |      |           |
| $m^2$ | 32.04 | mN/m | (sig P S) |
| $c$   | 5.45  |      |           |
| $c^2$ | 29.65 | mN/m | (sig D S) |

Finally, the surface energy being the some of the polar and dispersive fraction, we calculate of surface energy of  $32.04 + 29.65 = \mathbf{61.69 \text{ mN/m}}$  for the D263bio sample tested.

To calculate the uncertainty on this result, we use the square root of the maximum of the fit residuals. As an example, absolute value of the square root difference between the y-intercept and the value obtained for di-iodomethane:  $\sqrt{6.38 - 5.66} = 0.97$

The final value of the uncertainty is rounded to 1 mN/m at  $k=1$  (68 % confidence interval).

**The surface energy can be expressed as 61.69 mN/m with an uncertainty of 2 mN/m ( $k=2$ , 95 % confidence interval).**



## Conclusion

In this report, we demonstrated the experimental applicability of the wettability test protocol described in activity A3.2.2. We performed a documented example of an experimental assessment of the surface energy of a D263bio glass sample using the OWRK model tested with ultra pure water, di-iodomethane and ethylene glycol. The measurement system for contact angle was calibrated using traceable angle standards and a comparison of the accuracy of contact angle software tools is also provided in the report.